Spatial-Temporal Convolutional Networks for Skeleton-Based Dynamic Hand Gesture Recognition

Po-Yun Cheng [1]

Abstract

1. Introduction

Hand Gesture Recognition (HGR) has gained growing attention in recent years as image classification models have become increasingly powerful. This advancement makes HGR feasible on edge devices without the need for complex hardware such as depth or 3D cameras. One of the most common applications of HGR is desktop control, where it explore using hand gestures to control a computer desktop.

Convolutional Neural Networks (CNNs) are the most widely adopted technique in this domain, and many studies (Chua et al., 2022; Khatri et al., 2024; S K & Sinha, 2021) have explored various CNN variants to improve recognition accuracy and efficiency. However, from a human–computer interaction (HCI) perspective, most existing systems map static or dynamic gestures purely to discrete command triggers, often resulting in interaction techniques that are unintuitive and difficult to learn. In contrast, we propose a gesture set that combines continuous 2D finger pointing for cursor movement with discrete dynamic gestures for control actions, leveraging familiar metaphors from traditional input devices to provide a more natural and learnable interaction experience.

Recognizing dynamic gestures requires the ability to capture both spatial and temporal patterns. Prior work (Slama et al., 2025; Lea et al., 2016; Lu et al., 2018) has demonstrated that Temporal Convolutional Networks (TCNs) and Spatial Graph Convolutional Networks (S-GCNs) are effective for modeling such spatiotemporal features in action recognition tasks. Building upon these findings, we propose a real-time dynamic hand gesture recognition framework that combines GCN and TCN architectures. Our pipeline consists of four stages: (1) a MediaPipe Hand Landmarker extracts 3D hand landmark coordinates from each 2D camera frame, (2) a GCN encodes spatial hand-pose features, (3) a TCN captures temporal motion dynamics across frames, and (4) a classifier outputs one of eight gesture classes. Stages (2)–(4) collectively form the Spatial-Temporal Convolutional Network (STCN) used for gesture recognition.

[1]University of California Merced, USA. Correspondence to: Po-Yun Cheng <po-yuncheng@ucmerced.edu>.

2. Gesture Design

Figure 1 illustrates the set of gestures recognized by our system. The gestures are designed to be intuitive for users while intentionally avoiding reliance on depth information, making them easier to detect. The `press` and `release` gestures are inspired by mouse clicking, where the thumb acts as a "virtual" button. The `scroll` gestures follow the interaction style of Windows touchpad scrolling.

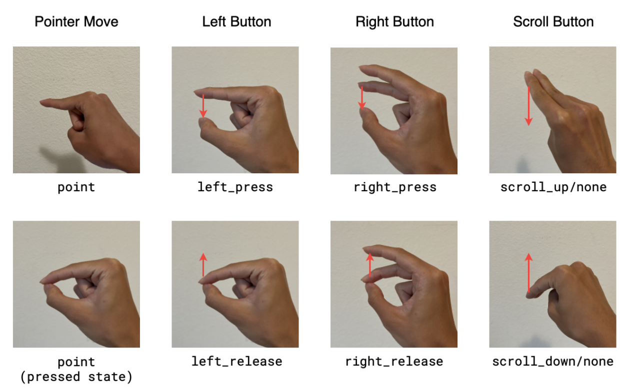

Figure 1: The gesture set consists of a static `point` gesture for controlling the pointer's 2D movement and six dynamic gestures representing button actions. Unrecognized gestures are labeled as `none`.

To be notice that, the same gesture may correspond to different actions depending on the context. For instance, during a scrolling-down interaction, a `scroll_up` movement is interpreted as `none`, as it simply serves to quickly reset the hand position in preparation for the next downward scroll.

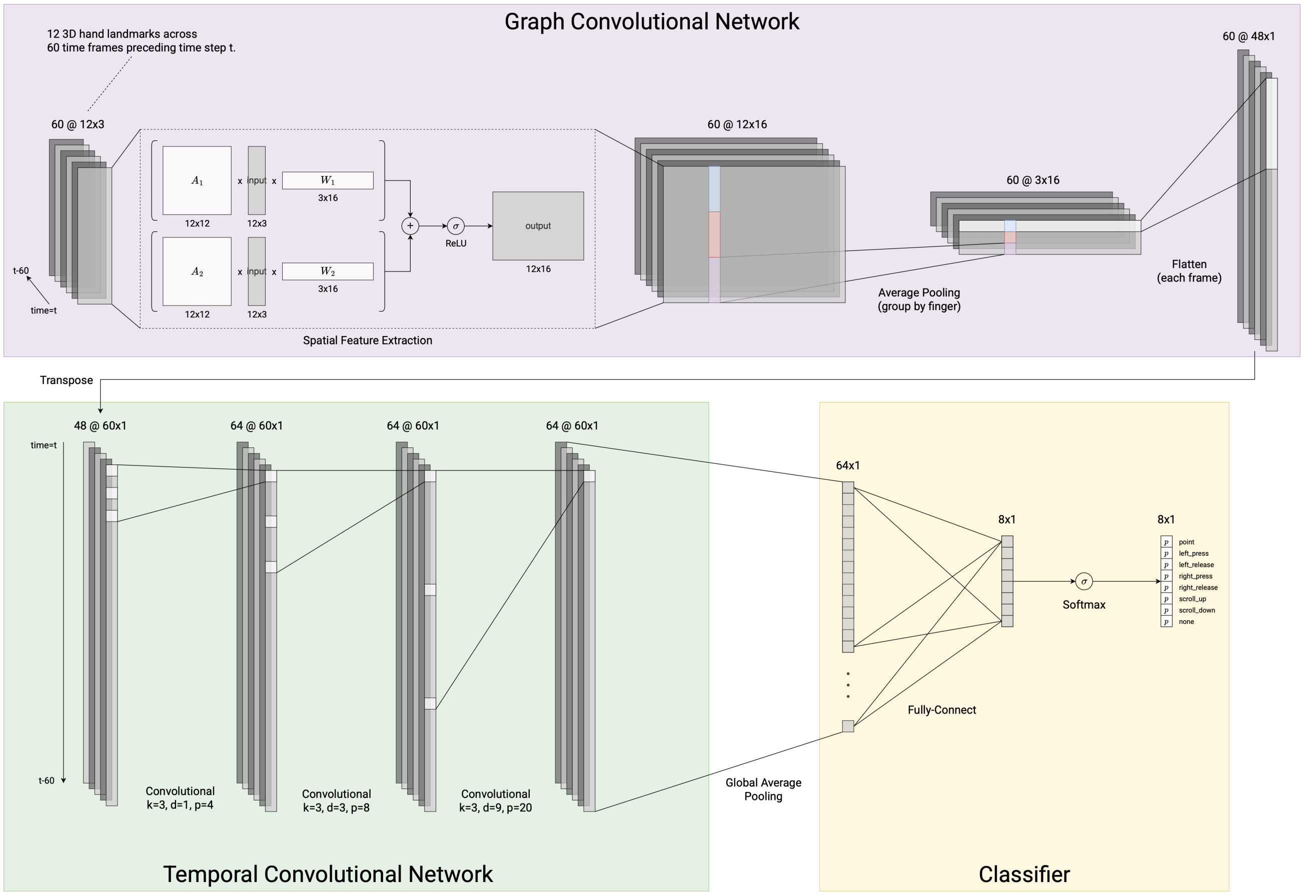

Figure 2: Architecture of the proposed STCN. The input is a sequence of detected hand landmarks, and the output is a probability distribution over eight gesture classes. Here, k denotes the kernel length, d the dilation size, and p the tail-padding size.

3. System Architecture

Figure 2 shows the architecture of the proposed STCN. It is worth noting that, in the current version, our model focuses only on the thumb, index, and middle fingers of the right hand to minimize the number of learnable parameters.

3.1. Graph Convolutional Network

The definition of a gesture depends on both the inside-finger spatial representation (the pose of each individual finger) and the inter-finger spatial relationships over time, with different gestures placing varying emphasis on each. For example, a `left_press` gesture can be interpreted as a straightened index finger while it gradually moves closer to the thumb over time. To capture both spatial perspectives, we construct two adjacency matrices, A_1 and A_2, to define finger skeletons, as illustrated in Figure 3.

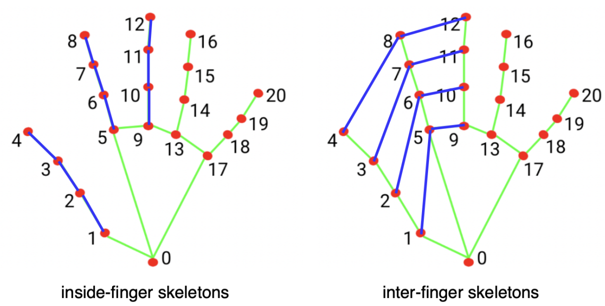

Figure 3: The blue lines indicate the inside-finger and inter-finger skeletal connections, defined by the adjacency matrices A_1 and A_2, respectively.

3.2. Temporal Convolutional Network

4. Methodologies

4.1. Training Data Collection

4.2. Evaluation

5. Experimental Results

6. Conclusion

References

Chua, S. N. D., Chin, K. Y. R., Lim, S. F., and Jain, P. Hand gesture control for human–computer interaction with deep learning. *Journal of Electrical Engineering Technology*, 17(3):1961–1970, January 2022. ISSN 2093-7423. doi: 10.1007/s42835-021-00972-6. URL http://dx.doi.org/10.1007/s42835-021-00972-6.

Khatri, S., Chourasia, L., Jain, A., and Barge, Y. Touchless control: Hand-based gesture recognition for human-computer interaction. In *2024 Parul International Conference on Engineering and Technology (PICET)*, pp. 1–6, 2024. doi: 10.1109/PICET60765.2024.10716196.

Lea, C., Vidal, R., Reiter, A., and Hager, G. D. Temporal convolutional networks: A unified approach to action segmentation, 2016. URL https://arxiv.org/abs/1608.08242.

Lu, D., Qiu, C., and Xiao, Y. Temporal convolutional neural network for gesture recognition. In *2018 IEEE/ACIS 17th International Conference on Computer and Information Science (ICIS)*, pp. 367–371, 2018. doi: 10.1109/ICIS.2018.8466467.

S K, S. and Sinha, N. Gestop: Customizable gesture control of computer systems. In *Proceedings of the 3rd ACM India Joint International Conference on Data Science & Management of Data (8th ACM IKDD CODS & 26th COMAD)*, CODS-COMAD '21, pp. 405–409, New York, NY, USA, 2021. Association for Computing Machinery. ISBN 9781450388177. doi: 10.1145/3430984.3430993. URL https://doi.org/10.1145/3430984.3430993.

Slama, R., Rabah, W., and Wannous, H. Online hand gesture recognition using continual graph transformers, 2025. URL https://arxiv.org/abs/2502.14939.